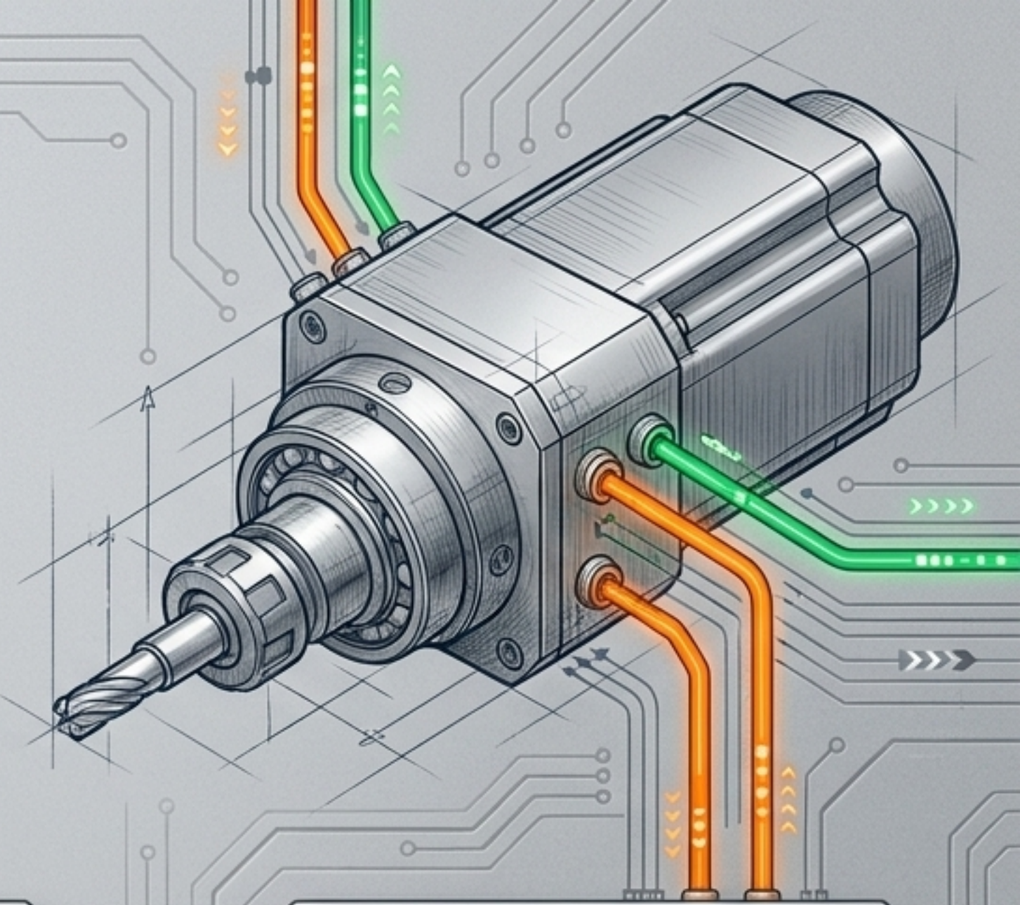
ME6324 — Artificial Intelligence in Manufacturing | Group 7

Hybrid AI-Expert System for Automated Manufacturing

[SYSTEM STATUS: ACTIVE]

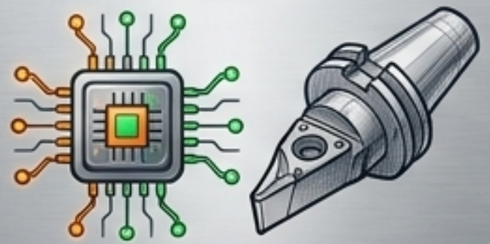
A rule-based + soft-computing pipeline for autonomous CNC decision support.



Process Planning (CAPP)



Cutting Tool Selection



Multi-Objective Machining Optimisation



Gemini Notebook

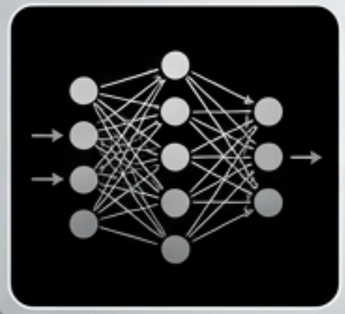
Problem & Motivation

1 Static Handbook



Manual, static lookup ignores real-time tool degradation.

2 Pure Machine Learning



Cannot infer operation sequences, logically select tools, or explain faults like chatter and chipping.

3 The Machining Reality



Flank wear (V_B) and surface roughness (R_a) evolve non-linearly.

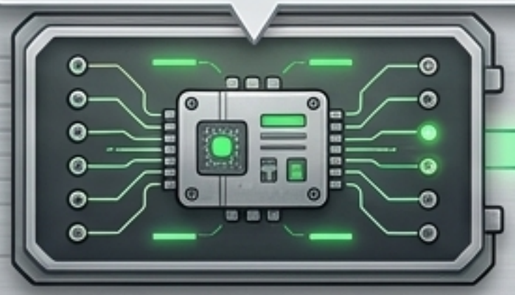
Maximising Material Removal Rate (MRR) directly conflicts with tool life and finish.

Manual, static planning cannot keep pace with real machining conditions — or resolve competing goals.

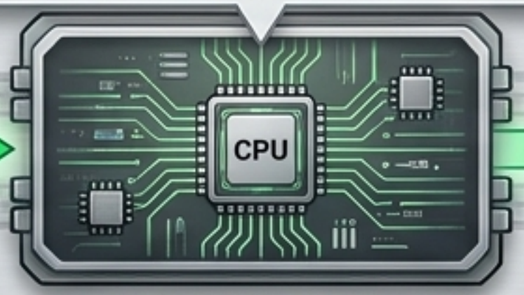
Cycle Time
Eliminates manual lookup & trial-and-error setup.

Scrap & Rework
Predictive wear + fuzzy monitoring flag bad tooling early.

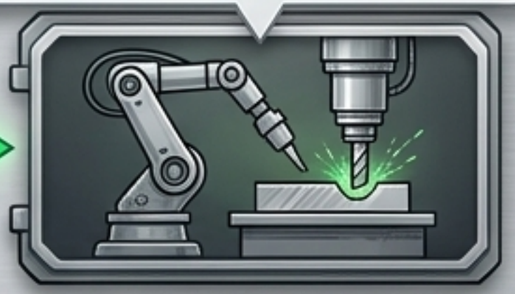
Tool Cost
Optimised parameters cut tool wear & energy use.



Sensor Data



Inference Engine



Actuation Recommendation

Gemini Notebook

Data Sources & Preprocessing

Two complementary, publicly available datasets cover sensor-based wear data and standardized tooling specs.

NASA Milling Dataset
(NASA Ames / UC Berkeley BEST Lab)

- 167 runs on a Matsuura MC-510V vertical machining centre.
- **Key Fields:** Cutting speed, feed, depth of cut, material; 6-channel sensors (current, vibration, AE); flank wear V_B.
- **Role:** Trains the ANN wear / roughness prediction model.

Sandvik Coromant
ISO 13399 Tooling Catalog

- Structured tool geometry & recommended speed/feed envelopes.
- **Key Fields:** Insert shapes (CNMG, WNMG, APMT), nose radius, rake angle; ISO groups P/M/K/N/S.
- **Role:** Feeds CLIPS rule-based tool selection.

Raw Sensor Signals -> FFT / RMS Feature Extraction -> Normalised Feature Vectors

Tooling Catalog -> Parsed & Structured -> SQLite Schema (ISO-keyed)

Preprocessing Pipeline



Gemini Notebook

Methodology — Four-Stage Hybrid AI Pipeline

Each AI paradigm handles the part of the problem it is best suited for — chained sequentially with feedback.





Part Specs

STAGE 1: CLIPS
Symbolic AI

STAGE 2: ANN
Neural Networks

STAGE 3: Fuzzy
Fuzzy Logic

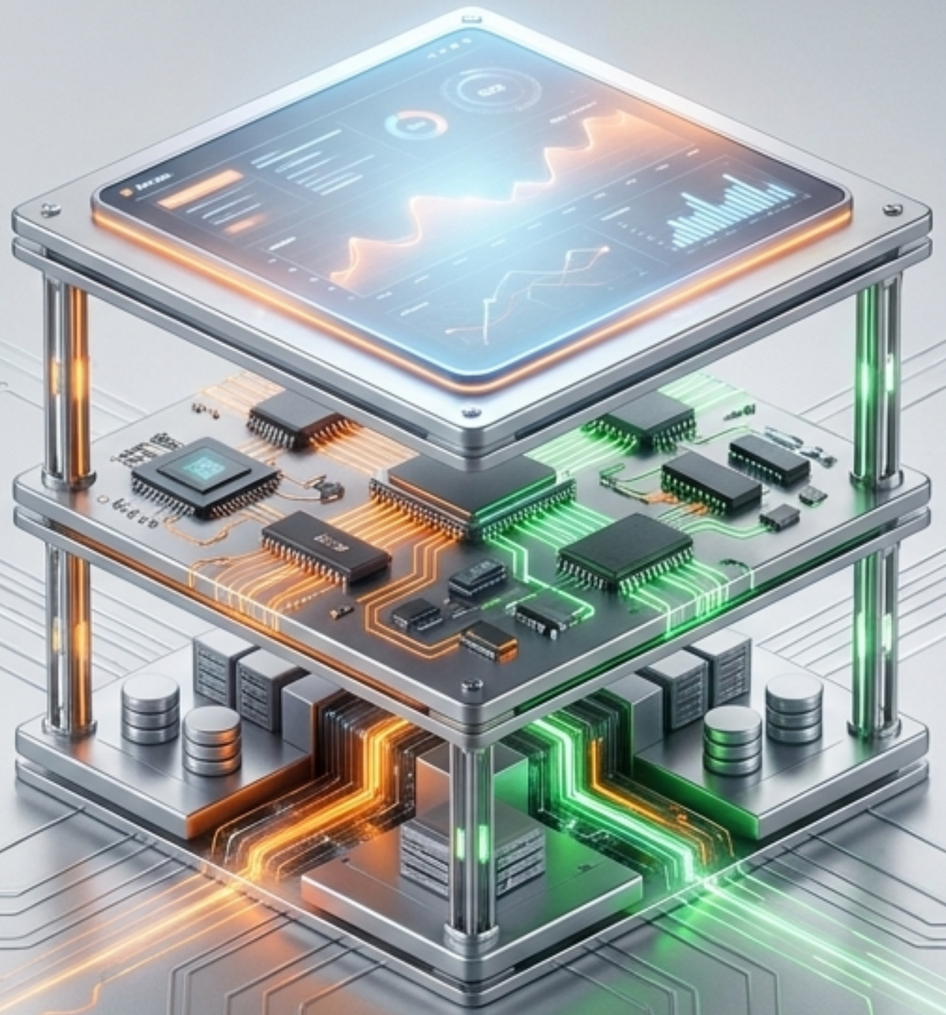
STAGE 4: NSGA-II
Evolutionary Algorithms

Interactive Dashboard

Feedback loop: diagnostic output from Stage 3/4 routes back to CLIPS for re-diagnosis and re-planning when faults occur.

Gemini Notebook

System Implementation & Tech Stack



Presentation Layer

Streamlit dashboard — CAPP sheet, 3D Pareto front (Plotly), real-time diagnostic explanations.

AI / Logic Layer

CLIPS via clipspy · PyTorch / Scikit-learn (ANN) · scikit-fuzzy · DEAP / pymoo (NSGA-II).

Data Layer

SQLite + SQLAlchemy · NASA sensor dataset · Sandvik ISO 13399 tooling catalog.

Fully open-source and free — modular architecture lets each of the 5 team members build and test their module independently, integrated through a single dashboard.

Gemini Notebook

Validation & Future Scope

Unifies Expert Systems, Neural Networks, Fuzzy Logic & Genetic Algorithms into one CAPP + optimisation + diagnosis pipeline — fully aligned with the ME6324 syllabus.

CLIPS

Rule coverage & correctness vs. handbook ground truth.

ANN

Low RMSE / MAE on held-out NASA test runs (V_B, R_a).

Fuzzy

Safe / Caution / Critical boundary accuracy.

NSGA-II

Pareto hypervolume & spread of non-dominated solutions.

System Speed

Dramatic planning-time reduction compared to manual lookup.

Physics Validation

Outputs validated against Taylor Tool Life physics checks.

1

Telemetry

Real-time CNC telemetry (OPC-UA / MTConnect) replacing static datasets.

2

Edge Compute

Edge deployment of ANN & fuzzy modules directly on CNC controllers.

3

Closed-Loop

Closed-loop adaptive control: NSGA-II output auto-adjusts feed/speed dynamically.

4

Expansion

Extend to new materials (Inconel, titanium) & multi-axis machining.

Gemini Notebook